Srevarshan Subramanian

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PROFESSIONAL SUMMARY

AI Engineer and final-year Computer Science student who builds complete software products, from system design to cloud deployment. I've integrated Generative AI into a production credentialing application for a US-based healthcare company, and built offline Edge AI devices for agriculture. I hold a published Government of India patent and am a national hackathon winner, recognised at IIT Madras with IIM product mentorship. I enjoy turning real-world problems into reliable, working solutions.

B.Tech – Computer Science Engineering (AI & ML) *2023 – 2027*

SRM Institute of Science and Technology **CGPA:** 9.75 / 10

TECHNICAL SKILLS

Languages & Fundamentals: Python, SQL, Data Structures & Algorithms, OOP, Problem Solving

Generative AI: RAG Pipelines, LangChain, LLM APIs, Vector DB (ChromaDB), Ollama, Prompt Engineering, Agentic AI Systems, LLM APIs & SDKs, VLM,OCR, Fine Tuning

AI / ML Frameworks: PyTorch, Hugging Face, Scikit-learn, Nvidia NGC, Numpy, Pandas

Workflow & Dev Tools: n8n workflow orchestration, Claude Code, OpenAI Codex, Antigravity

Backend & Deployment: FastAPI, REST APIs, Docker, AWS, Git, Github, PostgreSQL

EXPERIENCE

AI Product Developer | **EmedLogix** Dec 2025 – Jun 2026

- Automated the healthcare credentialing process for a US-based client by adding Generative AI features to a traditional credentialing application.
- Built **real-time OCR** and document parsing pipelines with automated data extraction and intelligent autofill, reducing manual data entry by ~90%.
- Integrated live **API-based data retrieval of US Govt End-points** and autofill systems to streamline multi-step onboarding workflows.
- Designed a natural language database agent using **AWS RDS PostgreSQL**, enabling conversational querying of structured relational data.
- Deployed the containerized backend on **AWS Elastic Beanstalk** and provisioned the frontend via **AWS Amplify** to ensure real-world enterprise scalability and reliability

AI/ML Intern | Microsoft × Edunet Foundation May 2025 – June 2025

- Completed Microsoft Learn's machine learning program, applying supervised learning algorithms to real-world problems.
- Chose **Random Forest** for my internship capstone and built **AI Nose**, a real-time environmental gas monitoring system using **ESP32**, **MQ3/MQ9 sensors**, and **Firebase**.
- Collected sensor data, trained the model to classify odor levels with **98% accuracy**, and streamed live readings to a web dashboard for real-time monitoring.
- Implemented automated sanitation alerts based on model predictions, demonstrating an end-to-end IoT and machine learning solution.

PROJECTS

Banana Weevil Early Detection System (Patent Published Research Project| **Edge AI** Jetson Orin Nano)

- Built a dual-modal acoustic sensing system combining passive bio-acoustic detection and active percussive resonance analysis to **identify internal weevil infestations** non-invasively.
- Designed an **edge AI pipeline** with MFCC feature extraction and MobileNetV2 CNN classifies infestation into 5 larval stages in real-time with nearly 80% accuracy, fully offline on AI device.
- Implemented end-to-end DL processing pipeline (**audio acquisition** → **filtering** → **MFCC** → **CNN inference** → **display**) on Raspberry Pi, optimizing real time inference on NVIDIA Jetson Orin Nano for lower latency.
- Estimated to cut banana crop yield loss from ~40% to less than 15% via early intervention, designed as a low-cost, cloud-free system for rural field usage.
- **Government Patent Published**: "System for Banana Weevil Detection using Dual-Mode Acoustic Sensing" (Govt. of India | App No. 202541116228 A)

NutriMinds AI – Personalized Nutrition Analysis (Springer ICDAM 2026 Publication | GenAI, VLM, Fine-tuned LLM)

- Built an end-to-end platform that reads a user's medical report and a packaged food label, then tells them whether the food is safe, recommended or not recommended for their health conditions.
- Designed a multimodal extraction pipeline the Vision-Language Model reads medical report tables and terminology, and PaddleOCR pulls multilingual ingredients, serving sizes and additive codes from food labels.
- Fine-tuned Mistral 7B Instruct with QLoRA (4-bit, Unsloth) on a dataset combining 900,000+ Open Food Facts products, clinical nutrition therapy rules for conditions like diabetes, kidney disease and hypertension, and synthetic disease-food scenarios.
- Served the model through FastAPI behind a React dashboard; each request returns a JSON result with a compatibility score, risk analysis (allergens, additives, sugar/sodium/potassium) and a safe serving guideline.

PUBLICATIONS

- **Research Paper (Accepted)** – *"Nutri-Mind: Personalized Nutritional Analysis Using Vision and Small Language Models"* – **Springer Proceedings, 7th International Conference on Data Analytics and Management (ICDAM 2026)**, organized by **London Metropolitan University** *(Top 20% of 2,000+ global submissions).*

PROFESSIONAL CERTIFICATIONS

- Getting Started with Artificial Intelligence – IBM
- Getting Started with AI on Jetson Nano – NVIDIA
- Supervised Machine Learning: Regression and Classification – DeepLearning.AI / Stanford (2024)
- Fundamentals of Deep Learning – NVIDIA (2024)
- Model Context Protocol: Advanced Topics - Anthropic

ACHIEVEMENTS & RECOGNITION

- **Government Patent Published** – "System for Banana Weevil Detection using Dual-Mode Acoustic Sensing" | App No. 202541116228 A | Govt. of India
- 1st Place – **Spectrum25 National level Hackathon (VIT Chennai)**
- 1st Place – **Euphoria '25 SolveX AI Challenge Hackathon at Kalasalingam University**
- Maintained 9.7+ CGPA consistently, ranked Top 1% of cohort at SRM Trichy
- Top 8 **National Finalist in NAF AGRO UZHAVU Hackathon at IIT Madras**